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READY FOR CRISIS?

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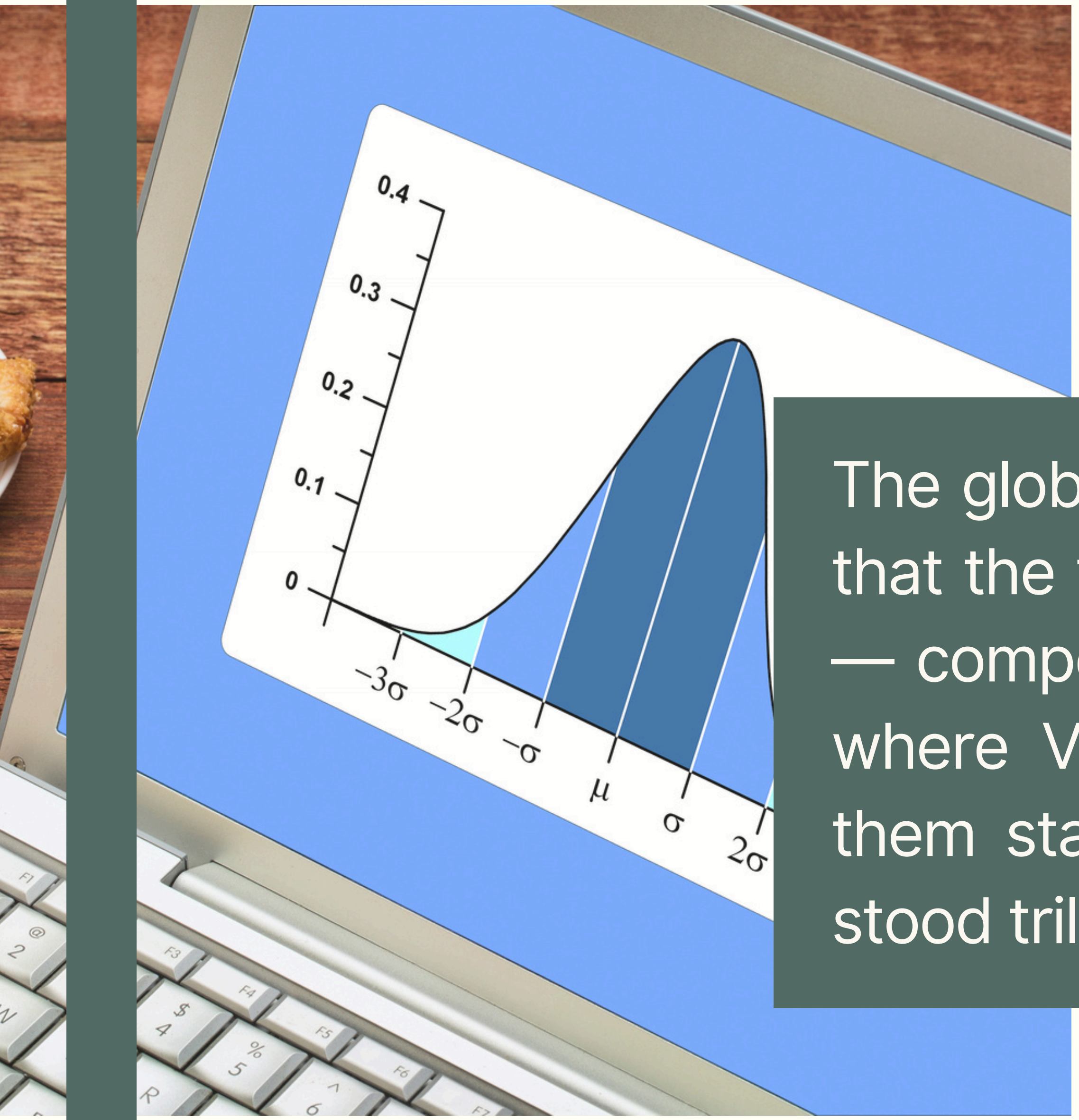
THE WORLD IS NORMAL



The global financial system before 2008 was built on the assumption that the total value of the world's assets follows a normal distribution — composed of millions of weakly correlated normal components — where Vasicek justified correlations economically and Li estimated them statistically, but both saw the same world and on that world stood trillions of dollars.

GLOBAL ASSUMPTION IN THE BEGINNING OF 21TH CENTURY

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THE MODEL EVERYONE BELIEVED IN

SIMPLE AND ELEGANT

This was one coherent image of the world. Mathematically elegant. It gave concrete numbers. Everyone used it.

SYSTEM COMPONENTS

Asset value of each borrower — Brownian motion with drift, normal shocks, parameters stable and slowly varying through GARCH. Every borrower has their own independent normal.

Correlation between borrowers — small, stable, described by Li's Gaussian copula. Arising only from local economic connections.

The system as a whole — the sum of millions of independent normals. CLT works. Risk is quantifiable, tails are thin, systemic catastrophe is impossible.

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THEY SAW A WORLD THAT WAS SIMPLE AND CALCULABLE

- NO UNCERTAINTY → RISK
- MATHEMATICAL → CALCULABLE
- SIMPLE → REDUCED
- SOPHISTICATED → PRESTIGIOUS
- NORMAL → NO UNEXPECTED SHOCKS
- INDEPENDENT → NO RISK AGGREGATES

This encourages greater courage. Since everything is simple ...

SECURITIZATION

GOAL: RISK DIVERSIFICATION

- **POOLING**

A single mortgage carries high risk — the borrower may default. But if you take 1000 loans from different regions, different borrowers, different types — then statistically only a fraction of them will fail. The rest will repay. The risk averages out.

- **RISK DIVERSIFICATION**

Credit risk instead of sitting in one bank — disperses across the entire financial system. Pension funds in Norway, banks in Germany, funds in Japan. Theoretically if something goes wrong — the pain is distributed, no node of the system collapses.

- **THE LOGIC OF TRANCHING**

The pool of loans was divided into tranches according to the order of losses:

- Equity tranche (first loss) — small, highly risky, high coupon
- Mezzanine tranche — medium risk
- Senior tranche — last to lose, AAA rating, low coupon

- **TRANCHING: IDEA**

The idea: the investor chooses how much risk they want. A conservative pension fund buys AAA, a hedge fund buys equity at a high return. Everyone gets what they want.

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SECURITIZATION

INSTRUMENTS

- **MBS**

MBS (Mortgage-Backed Security) — A security backed by a pool of mortgage loans. Investors receive payments from the underlying borrowers' monthly repayments of principal and interest.

- **CDO**

CDO (Collateralized Debt Obligation) — A structured security backed by a pool of debt instruments, divided into tranches with different risk and return profiles. Senior tranches receive payment first and carry AAA ratings, while equity tranches absorb losses first and offer higher yields.

- **CDS**

CDS (Credit Default Swap) — A contract in which one party pays a periodic premium to another in exchange for protection against the default of a reference entity. It functions as insurance against credit risk, allowing risk to be transferred without selling the underlying asset.

- **SYNTHETIC CDO**

Synthetic CDO — A CDO that does not hold actual loans or bonds, but instead gains exposure to credit risk through credit default swaps. It allowed the creation of unlimited CDO tranches referencing the same underlying assets, massively amplifying the system's total exposure.

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SCALE OF SECURITIZATION

- **CDO**

Total value of CDOs in circulation 2007 — approximately 2 trillion dollars.
Of which:

- CDOs based primarily on subprime MBS — approximately 600 billion
- CDOs based on corporate and leveraged loans — approximately 1.4 trillion

- **CDS**

Estimated real net exposure — approximately 2-3 trillion dollars.

- **SYNTHETIC CDO**

Value of synthetic CDOs in circulation 2007 — difficult to estimate, approximately 1-2 trillion dollars.
Difficult to estimate because synthetic CDOs were deliberately opaque and often off-balance-sheet.

- **REPO**

All of the above instruments were financed through short-term repo. Daily turnover of the US repo market in 2007 — approximately 4-5 trillion dollars.

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SPECULATIVE ARBITRAGE

• BASEL I (1988) → RESPONSE

8% capital requirement on RWA.
Immediate effect:
Banks saw that holding loans on the balance sheet costs capital. Every 100 dollars of mortgage loans froze 4 dollars of capital that was not earning. A simple calculation emerged — if I can move the loan off the balance sheet, I free up capital and can issue a new loan.

• FOR NOW CAUTIOUSLY

Securitization began to grow systematically but still moderately. Commercial banks began securitizing mainly mortgage loans — because they carried a 50% weight and a 4% capital requirement. It was profitable to move them off the balance sheet. The CDO market began to develop in the 1990s — the first structures were relatively simple, mainly corporate bonds and leveraged loans.

• BASEL II (2004) - EXPLOSION

First — Advanced IRB and internal models.
Large banks could now use their own models to calculate RWA. Banks that received regulatory approval for Advanced IRB began massively reducing their RWA through "more precise" modeling. The effect was immediate and dramatic. Some banks reduced their RWA by 30-50% by switching to internal models — without any change to the actual loan portfolio. This meant that capital that was frozen — was suddenly released. Without any securitization. The model alone did the work.

Second — external ratings in the Standardized Approach.
AAA instrument — 20% risk weight. BBB instrument — 100% risk weight. Unrated instrument — 100% risk weight. This created direct demand for AAA-rated instruments. A senior CDO tranche with an AAA rating allowed banks to hold five times less capital than a regular loan.

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ENGINES OF EXPANSION

ENGINES

Engine one — internal models release capital that needs to be invested somewhere. Banks search for assets with high returns and low risk weights according to the model.

Engine two — AAA rating gives capital relief. A senior CDO tranche with AAA is five times cheaper in capital terms than a regular loan. CDOs were the answer to both engines simultaneously — high coupons with low risk weight thanks to the AAA rating.

THE CORE

Basel I created the motivation for securitization — but the market needed a decade to build the infrastructure.
Basel II hit a fully built securitization infrastructure and added fuel through two new mechanisms — internal models and the capital premium for AAA ratings.
The effect was like pouring gasoline on a smoldering fire. A system that had grown for a decade at a moderate pace — exploded within three years.

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EFFECTS OF ARBITRAGE

	Basel I (1988)	Basel II (2004)
Escape mechanizm	Securitization — move loans off the balance sheet	Internal models — reduce RWA without moving assets
Additional effect	None	AAA rating = capital relief → demand for CDO
Scale of response	Moderate, gradual	Explosive, 5-10x within a few years
New instruments	MBS, first CDOs	CDO squared, synthetic CDO, CDS
Hidden leverage	Moderate	Astronomical

REPO

WHAT IS REPO?

Repo — repurchase agreement — is an overnight loan. A bank sells assets with an obligation to repurchase them the next day at a slightly higher price. The difference is the cost of the loan.

INVESTMENTS BANK

Investment banks like Lehman financed themselves almost exclusively through repo — not through retail deposits. Every day they borrowed hundreds of billions of dollars overnight using CDOs and MBS as collateral. They transformed long-term assets — CDOs with twenty-year maturities — into overnight obligations. This worked as long as the repo market was open and as long as the collateral had an acceptable value.

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AIG - INSURANCE

AMERICAN INTERNATIONAL GROUP

American International Group — was the largest insurer in the world. It had one division — AIG Financial Products — which sold credit protection through CDS on CDO tranches. Investment banks built CDOs and bought insurance from AIG on senior tranches — those with AAA ratings. They paid AIG a small premium. AIG committed to pay if the tranche lost value. For banks this was ideal — they transferred risk off the balance sheet and released regulatory capital. **For AIG the model seemed great — Li's copula said that the probability of loss on an AAA tranche was close to zero**. AIG collected premiums on 80 billion dollars of exposure while holding almost no reserves.

AIG WAS SYSTEMICALLY CRITICAL

AIG was the hidden center of the entire system. When banks showed regulators that they had low risk — because it was insured by AIG — all that comfort depended on AIG's solvency.

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LEVERAGE

OFFICIAL VS. REAL LEVERAGE

Officially banks reported leverage in line with Basel II — it looked relatively safe.
But real leverage was dramatically higher because one had to add:
Off-balance-sheet SPVs and SIVs — structures that were formally not the bank but the bank had credit lines to them. When an SPV had problems — it came back to the bank. The risk was the bank's, just hidden.
CDS and derivative instruments — the notional value of contracts was a multiple of balance sheet assets. AIG had CDS outstanding with a notional value of approximately 500 billion dollars against capital of a few tens of billions.
Repo and short-term funding — investment banks financed long-term assets with short-term repo loans, renewed daily or weekly. This was hidden liquidity leverage.

LEVERAGE - BEFORE 2008

Lehman Brothers: 30:1 ⟶ 40-50:1
Bear Stearns: 33:1 ⟶ 40:1
Merill Lynch: 28:1 ⟶ 35:1
Deutsche Bank: 50:1 ⟶ unknown
AIG (with CDS): safe ⟶ 100:1 +

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RISK CONCENTRATION

- **CDO CONCENTRATION**

The CDO market was dominated by perhaps 10-15 banks that created the vast majority of instruments. These banks together issued perhaps 80-90% of all CDOs in the world.

- **RELATION**

Banks did not only create CDOs — they bought CDOs from each other. A system that looked like a distribution of risk to external investors — was in reality a circulation of risk between the same institutions. The risk never left the banking system. It circulated in a loop between the same 10-15 banks in increasingly complex packaging.

- **AIG - CONCENTRATION**

AIG sold CDS to virtually all major banks simultaneously. Everyone thought they were diversified because they held CDOs from different banks secured by CDS. But all CDS led to one counterparty — AIG. Instead of counterparty risk diversification — total concentration in a single node.

- **THE SAME UNDERLYING ASSETS**

Ultimately everything came down to one market — US real estate.
Subprime MBS — backed by US real estate.
CDOs — backed by subprime MBS.
Synthetic CDOs — bets on CDOs backed by subprime MBS.
CDS — insurance on CDOs backed by subprime MBS.

The entire pyramid of financial instruments worth tens of trillions of dollars stood on one foundation — US real estate prices.

REAL MONEY IN BANKS – 2007

Bank	**Assets (trillions USD)**
Citigroup	2.2
Deutsche Bank	2.0
UBS	2.0
BNP Paribas	2.0
Barclays	1.9
HSBC	1.9
Royal Bank of Scotland	1.7
Credit Agricole	1.6
JP Morgan	1.6

All together: 18 trillions, while World's GDP = 60 trillions (2007)

MODEL CONCENTRATION

- **LI AND GAUSIAN COPULA**

Li and the Gaussian copula — the greatest concentration. The Li copula was the standard for synthetic CDOs and CDO^2 — with an estimated notional value of 1-2 trillion dollars in CDOs plus 10-15 trillion in CDS on credit portfolios.

- **CREDIT METRICS**

CreditMetrics and its derivatives — the second level. CreditMetrics was the standard for traditional CDOs of corporate bonds and leveraged loans — approximately 800 billion to 1 trillion dollars in total.

- **KMV**

KMV — third level. KMV by Moody's was used for rating financial firms and banks, giving Lehman Brothers a high Distance to Default as late as July 2008 while maintaining an A2 rating — two months before its collapse.

- **CREDIT RISK +**

CreditRisk+ — fourth level. CreditRisk+ was used mainly by European banks — particularly German Landesbanks and Swiss banks — to price corporate loan portfolios worth several hundred billion dollars. Losses were painful but not systemic — Germany and Switzerland suffered, but were not the epicenter of the crisis.

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COPULA LI

MECHANIZM OF EXPANSION

• MECHANIZM

- Investment banks built synthetic CDOs.
- They used the Li copula to show rating agencies that the senior tranche was safe — AAA.
- Rating agencies — Moody's, S&P, Fitch — themselves used variants of the Li copula or accepted banks' calculations based on Li.
- AIG sold CDS on the senior tranches of these CDOs — insuring them. It used Li to price this risk as small.
- Investors — pension funds, European banks, sovereign funds — bought AAA tranches trusting the ratings.

• ALL USED LI LOGIC

The entire chain — investment bank, rating agency, insurer, investor — used the same Li logic. The same Gaussian copula. The same assumptions about zero tail dependence.

When mortgage default correlations exploded in 2007 — the entire chain broke simultaneously. Because everyone stood on the same foundation.

This is a systemic model failure. Not the failure of one institution — a failure that was pervasive because the model was pervasive.

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CRISIS

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THE MECHANISM OF COLLAPSE

1. **Trigger: real estate prices stopped rising**
The entire subprime system rested on one unspoken assumption — that even if a borrower could not repay, the property was worth more than the loan. When prices stopped rising in 2006, refinancing became impossible and borrowers began defaulting en masse.

2. **2006 — the first cracks.** Subprime delinquency indicators began rising but the market did not yet panic, treating early signals as isolated problems rather than systemic warning signs.

3. **February 2007 — HSBC**. HSBC announced subprime losses 20% higher than expected — the first major bank to say out loud "we have a problem." Markets trembled briefly but dismissed it as one bank's issue.

4. J**une 2007 — Bear Stearns funds.** Two Bear Stearns hedge funds invested in subprime CDOs announced their value had fallen nearly to zero, forcing Bear Stearns to rescue them with its own capital. For the first time the market saw that an AAA-rated CDO could be worth nothing — and rating agencies began mass downgrades, stripping hundreds of instruments of their AAA ratings within weeks.

5. **August 2007 — interbank market freezes**. BNP Paribas suspended withdrawals from three funds because it could not value their assets — not that they were worth nothing, simply that no one knew what they were worth. Every bank looked at every other bank and thought "I don't know how many toxic assets you hold — I won't lend to you." LIBOR spiked and the interbank market effectively froze.

6. **March 2008 — Bear Stearns collapses**. Bear Stearns borrowed short and invested long — when the interbank market froze it could not refinance and lost liquidity within days. The Fed and JP Morgan organized a last-minute takeover at 2 dollars per share, down from 170. Markets breathed — "the Fed rescues, the system is safe."

7. **September 2008 — Lehman Brothers.** On September 15, 2008 Lehman filed for bankruptcy after the Fed and US government decided not to rescue it. Within hours bank stocks collapsed globally, the commercial paper market froze, money market funds began "breaking the buck," and AIG — which had sold CDS on CDOs to virtually every major bank — was on the edge of collapse the following day, requiring an 85 billion dollar Fed rescue.

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WHEN THE WORLD OF VALUE COLLAPSES IN A BANK...

ASSETS < LIABILITIES

The bank loses because the valuation of assets is lower. Why? The invisible component of value has changed — the risk embedded within it.

If this loss of value exceeds the reserve capital — determined with a certainty of 1/1000 years :) — then the bank is in serious trouble. Its assets no longer cover its liabilities.

BANKING MODELS ARE SHOCKED!!!

BUNK RUN

Depositors fear the bank is insolvent and rush to withdraw their money simultaneously. The bank, holding only a fraction of deposits in cash, cannot meet all withdrawal demands at once. The very act of running on the bank causes the collapse that depositors feared — a self-fulfilling prophecy.

What is interesting is that at the crucial moment, they not only experienced a loss of value — they did not know what value their assets had

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AIG CENTRAL POINT

When CDOs began losing value in 2007 — counterparties demanded collateral from AIG. AIG suddenly had to post billions of dollars of collateral it did not have. A system that looked diversified — hundreds of banks, thousands of CDOs — was in reality concentrated in one place. When AIG was failing — the insurance of the entire system was failing simultaneously. The US government put up 180 billion dollars not to save AIG — but to prevent the simultaneous collapse of all the banks that thought they were insured.

CRISIS – REAL WORLD

- ## DESTRUCTION OF BANK CAPITAL

Banks lost 2 trillion dollars of equity capital.

At leverage of 10:1 — every dollar of lost capital means 10 dollars less in loans the bank can extend.

2 trillion in losses × leverage 10 = 20 trillion dollars less credit for firms and people.

Companies that normally renewed credit lines — suddenly could not. Not because they were poor credit risks — but because the bank had no capital to lend.

- ## CREDIT CRUNCH

When banks stopped lending — the real economy felt it immediately.

Small and medium enterprises — live on working capital credit. When a bank refuses to renew a credit line — the company cannot pay suppliers, cannot pay wages. It fails not because it has a bad business — but because it has no liquidity.

Developers — construction projects halted midway because the bank froze financing. Construction workers without jobs.

Consumers — banks tightened mortgage and auto loan criteria. US car sales in 2009 fell by 40% — not because people stopped wanting cars but because they could not get credit.

CRISIS - REAL WORLD

- **WEALTH EFFECT**

Stocks lost 30-50% of value globally. US real estate fell on average by 30%.

Millions of people whose life savings were in pension funds and stocks — suddenly saw they had 30-50% less.

The wealth effect — when you feel poorer you spend less, even if you have a job and income. This immediately hits consumption.
Consumption is 60-70% of GDP in developed countries. When it falls — GDP falls.

- **UNEMPLOYMENT**

Credit crunch + wealth effect + falling consumption = firms lose revenue = they lay off workers.

USA:
Unemployment in 2007 — 4.6%. Peak unemployment 2009 — 10%. Jobs lost — 8.7 million in 18 months.

Europe:
Eurozone — unemployment rose to 12%. Spain — unemployment reached 25% in 2012. Greece — unemployment 27% in 2013.

Every lost job is a family that stops paying its mortgage. Which generates further losses for banks. Which deepens the credit crunch. Another feedback loop.

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CRISIS – REAL WORLD

- **FISCAL CRISES AND BUDGET CUTS**

Governments had to rescue banks and simultaneously stimulate the economy.
USA: TARP — 700 billion bank bailout. Obama stimulus package — 787 billion. Budget deficit jumped from 1.2% of GDP to 9.8% of GDP in 2009.

UK: Nationalization of Royal Bank of Scotland and Lloyds. Deficit jumped to 11% of GDP.

Ireland: Total cost of bank rescue — 64 billion euros. With Irish GDP of 170 billion — this was 38% of annual GDP spent on rescuing banks. Ireland had to accept a bailout from the EU and IMF.

Greece: Banking crisis merged with debt crisis. Budget cuts for a decade. Greek GDP fell by 25% between 2008 and 2013 — a scale comparable to the Great Depression of the 1930s.

- **VIRTUAL LOSSES → REAL WORLD**

The financial system is not a separate world from the real economy. It is its circulatory system.

Credit is the blood that flows through the economy. Firms need credit to produce. People need credit to buy homes and cars. Governments need markets to finance roads and hospitals.

The 2008 crisis was a heart attack of the financial system. The consequences for the real body — the economy — were inevitable. Millions of people lost their jobs, homes, and life savings — not because they did anything wrong. But because a dozen banks built a pyramid of financial instruments on false assumptions and when the pyramid collapsed — it took the entire system with it.

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3 NUMBERS

- **10 MILLION**

Americans who lost their homes to foreclosure as the subprime pyramid collapsed.

- **26%**

the fall in Greek GDP between 2008 and 2013, deeper than the decline the United States experienced during the Great Depression of the 1930s.

- **15 YEARS**

the time it took for the median US household income to return to its pre-crisis level.

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5 NUMBERS

- **61 TRILLION DOLLARS**

The notional value of the CDS market at its peak in 2007 exceeded the entire global GDP of that year — the world placed credit insurance bets larger than all global production combined, on instruments priced by the Li copula.

- **180 BILLION DOLLARS**

The cost of the US government rescue of AIG — a single insurer that had sold CDS protection on CDOs priced by a model assuming zero tail dependence.

- **8 DAYS**

The time between the bankruptcy of Lehman Brothers on September 15, 2008 and the freezing of global credit markets — eight days for Danielsson's synchronization spiral to travel through the entire system.

- **1 DOLLAR**

The price of Citigroup stock in February 2009, down from 55 dollars in 2007 — a 98% collapse in eighteen months for a bank that still carried an AA rating just two years earlier.

- **0**

The number of Wall Street bankers who went to prison in the United States for causing the crisis.